

## Lesson 12 — If / Then / Else & Tracing Logic

Matches LEGO Teacher Portal lesson: "Level Up".

**Objective (I can...):** I can use if/then/else for two-way branching and trace which branch runs for a given input. **CS concept:** Two-way selection; tracing boolean logic · **Builds on:** Lesson 11 (if/then).

**AI Connection (2 min):** Face unlock branches: IF recognized, THEN unlock; ELSE stay locked and log the attempt. *Impacts prompt:* "Fast unlock helps most people — who could it lock out unfairly, and why?"

**Materials:** LEGO CS & AI kit + device per pair, trigger objects / color cards, notebooks. **Model reuse:** Conditionals model from Lesson 11 — code only.

**60-minute flow - Warm-up (8):** "Which branch?" — coach reads an if/then/else and an input value; pairs call the branch and the result. Model tracing on the board. - **Concept + demo (10):** Demo if/then/else: object near → green light + forward; else → red light + wait. Run both branches and trace each. (*Rotate format:* "trace it on paper first.") - **Build/Code (30):** Pairs code an if/then/else with two clearly distinct outcomes from one condition, then trace, in the notebook, which branch fires for each of two test inputs. *Swap at 15 min.* - **Challenge (7):** Make each branch a two-step action (light + motion) and re-trace. - **Share + pack-up (5):** Pairs demo both branches and state which condition selected which. Model stays built.

**Pair roles:** Driver codes the "then," Navigator designs and traces the "else"; swap at 15 min so each owns a branch. (*Trio: third runs both test inputs.*)

**Differentiation:** *Support* — the "then" provided; students add the "else" from framed pseudocode. *Extension / fast-finisher* — nest an if inside one branch. *Reteach* — trigger each branch by hand, label it, then code it.

**Exit check:** Pair's model shows two distinct outcomes correctly selected by one condition, and they can trace which branch runs for a given input. **At-home (optional):** Write an if/then/else for a real decision and trace it for two different situations. ~10 min.

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